

Hiver SDE Intern — Take-Home Assignment

What we are testing: whether you can turn a messy real-world dataset into a working AI system *and prove it works*. The proof is worth more than the system.

The problem

You are given real customer-support conversations between customers and brands on Twitter. Pick **one brand** from the dataset and build an AI support agent for it that can:

1. **Classify** each incoming customer message into a small set of intents that *you* define from the data.
2. **Draft a reply** grounded in how that brand has historically resolved similar issues.
3. **Decide** whether the message should be auto-handled or escalated to a human — with a stated reason.

You then have to convince us the agent is good enough to trust. That is the hard part.

Dataset

- **Primary:** *Customer Support on Twitter* (Kaggle, [thoughtvector/customer-support-on-twitter](#)) — ~3M tweets, multi-turn threads, dozens of brands. Real, noisy, and imperfect.
- **Optional secondary (for intent work only):** *Banking77* (Hugging Face [PolyAI/banking77](#)) — 13k queries, 77 labelled intents.
- You may use any LLM API or open model.

Deliverables

1. **Repo** with a runnable pipeline ([README](#) must let us reproduce your headline results in under 15 minutes).
2. **Golden evaluation set** — 150–250 hand-labelled examples you built yourself, with a short note on how you sampled and labelled them.
3. **Evaluation harness** — automated metrics + an LLM-as-judge rubric for reply quality, *including evidence of how well your judge agrees with a human*.
4. **Report (max 6 pages / or a README section)** covering:
 - Problem framing: what "good" means for this brand, and what you chose *not* to build.
 - Results vs. at least two baselines (a trivial one and a simple one).
 - Failure analysis: your top 5 failure modes with real examples and hypotheses.
 - "What is misleading about my headline number?" — a mandatory section.

- What you'd do next with one more week.
5. **Decision log** — a plain list of the 10–15 non-obvious decisions you made and why. Bullet points are fine.

How to submit

- Submit everything through this form:
<https://intelligent-bar-256.notion.site/39492cbf0da2800682cfc78a600a745f?pvs=105>
- Include the repo link (public, or private with access granted to us) and the report. Do not email submissions.

Rules

- You may use AI coding assistants freely. We will ask you to explain and modify your own code live.
- Cite anything you borrowed. Borrowing is fine; not knowing what you borrowed is not.
- We will not run your code on the full dataset — a subsample is expected and encouraged.